

	Work Instruction	Doc. Revision	01
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This procedure is applicable to the following models:

Machine Model	Server Option		
	S1	S2	S3
V810 / V810i STD			
V810 / V810i XXL		√	√
V810 / V810i S2EX		√	√
V810 / V810i Mini			
V810 / V810i XLT / XLTM / XLL		√	√
V810 / V810i XLW			
V810 / V810i S3		√	√

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	15 Aug 2024	First Release	Yeap Leong Wei
01	30 Oct 2024	Include XXL for applicable model + content update	Yeap Leong Wei

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Scope

This procedure provides guidelines for the removal and replacing the Y-Axis Quantic Encoder Read Head.

Tool List

No	Part No	Description	Qty	UOM	Images
1	-	Ball Point Metric Allen Wrench Set	1	SET	
2	-	Metric T-Handle Wrench (M5 size)	1	PC	
3	-	Camera Tool	1	PC	
4	-	Tamper Proof Screw Driver Set	1	Set	
5	-	Cutter	1	PC	

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6	4XASPP-A440	Renishaw Quantic Encoder Alignment Kit	1	Set	
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Table 1: Tool List

Upgrade Kit List

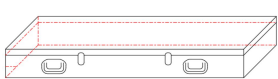
No	Part No	Description	Qty	UOM	Images
1	4XASPP-A425	RGH22 TO QUANTIC Y-AXIS ENCODER KIT (S2EX)	1	SET	
2	4XASPP-A427	RGH22 TO QUANTIC Y-AXIS ENCODER KIT (XLT)	1	SET	
3	4XASPP-A429	RGH22 TO QUANTIC Y-AXIS ENCODER KIT (S3)	1	SET	

Table 2: Upgrade Kit List

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Origin Y-Axis Encoder Read Head Removal Procedure

1. **Board Unloading:** Prior to commencing the removal procedure, ensure that any boards within the machine are safely unloaded. Follow the manufacturer's guidelines or equipment manual for proper board removal procedures.
2. **Shutdown Sequence:** Close V810 GUI and initiate the shutdown sequence for the Super Micro PC by following the manufacturer's recommended procedure. Ensure all data is saved and any necessary backups are performed before proceeding.
3. **Deactivation of Equipment:** At the operator console panel, press the “X-Ray Off” button to deactivate any X-ray equipment. Turn the key to the “Off” position to ensure complete deactivation of the system and associated components.
4. **Power Down Procedure:** Safely power down the SSCA main power source by following the manufacturer's recommended shutdown procedure. Ensure all power indicators are off before proceeding with the removal process.
5. **Access Door Opening:** Open all front access doors carefully to provide sufficient access to the internal components of the machine. Ensure doors are fully opened and secured to prevent accidental closure during the removal process.
6. **Left Outer Barrier Removal:** Remove the left outer barrier by unscrewing the door screws using the appropriate tools. Store the screws in a secure location for reassembly after the removal process is complete.
 - a) For S2EX & XXL/XLT



Figure 1: Left Outer Barrier (S2EX & XXL/XLT)

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b) For S3



Figure 2: Left Outer Barrier (S3)

7. **Rear Access Door Opening:** Open the rear access door carefully, ensuring there are no cables or obstructions that may hinder the removal process. Take caution to prevent strain on any cables or connectors located near the access door.
8. **Removal of the vertical lead cover:** Unfasten the screws and remove the vertical lead cover. Take caution to prevent damage to the lead cover by placing it in flat and cleaned surface.

a) For S2EX, XXL/XLT

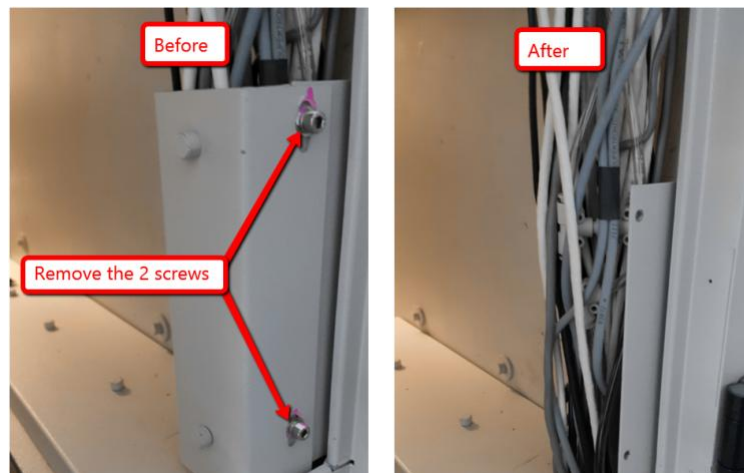


Figure 3: Vertical Lead Cover (Sample Image)

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b) For S3

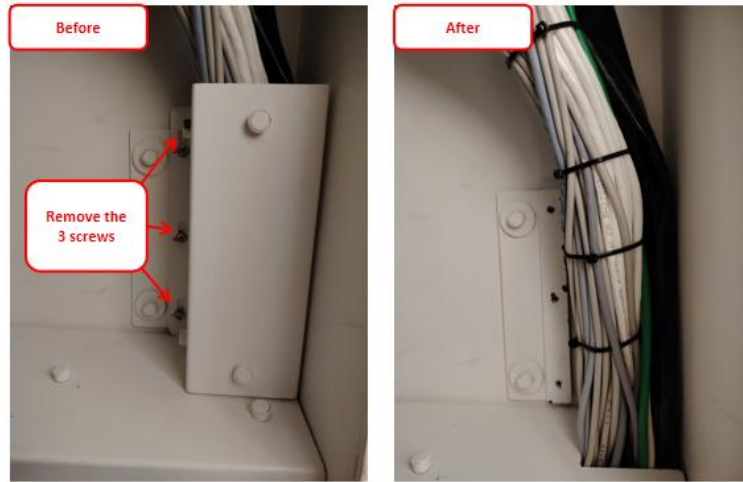


Figure 4: Vertical Lead Cover

9. **Removal of the horizontal lead cover:** Unfasten 3 screws and remove the horizontal lead cover. Take caution to prevent damage to the lead cover by placing it in flat and cleaned surface.

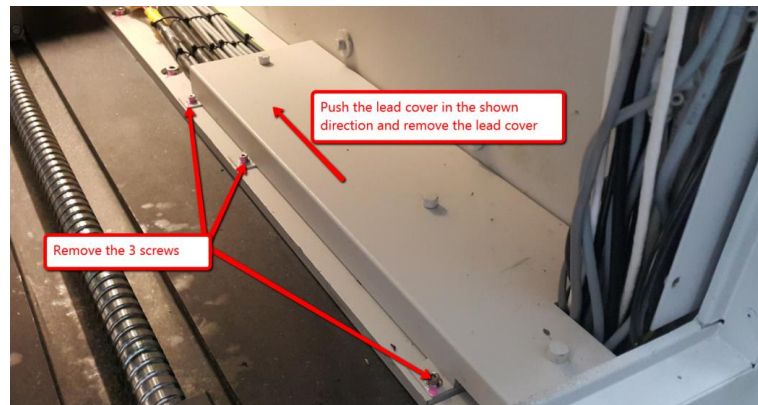


Figure 5: Horizontal Lead Cover (Sample Image)

10. **Removal of the SSCA Cable Bracket or Cable Tray Cover:** Remove the bracket or cable tray cover for the facilitation of the cable removal process.

a) For S2EX and XXL/XLT

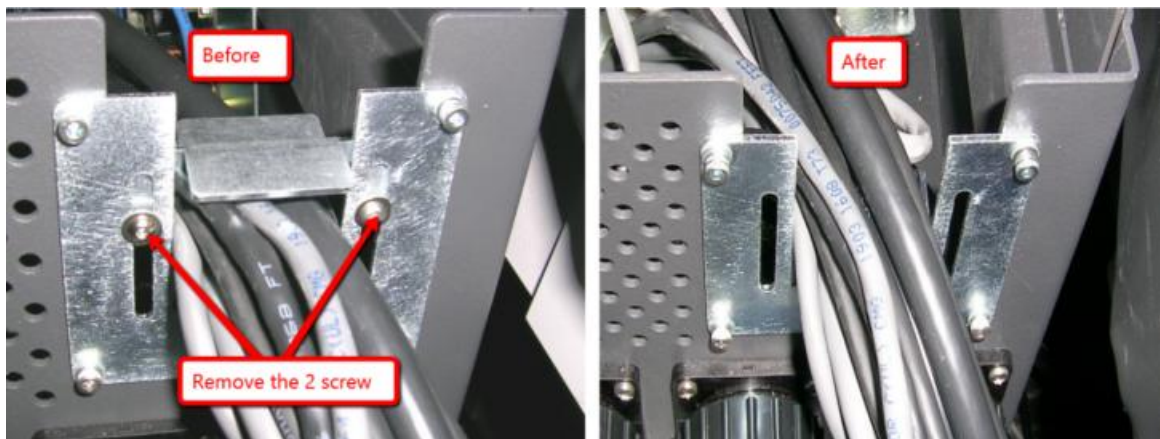


Figure 6: SSCA (Sample Image)

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b) For S3

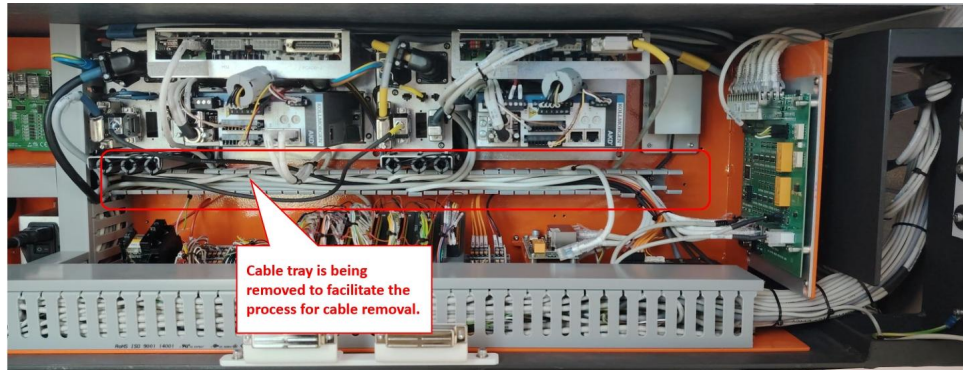


Figure 7:SSCA for S3 (Sample Image)

11. **Disconnected the Y-Axis Encoder Head Connector:** Unplug Y-Axis read head encoder connector from Y-Axis Kollmorgen kit (for S2EX) or from the extension cable connector (for XXL/XLT and S3).

Note: Remove necessary cable ties and carefully remove the encoder cable.

a) For S2EX and XXL/XLT

Note: For XXL/XLT, please unplug encoder read head from the extension cable connector.

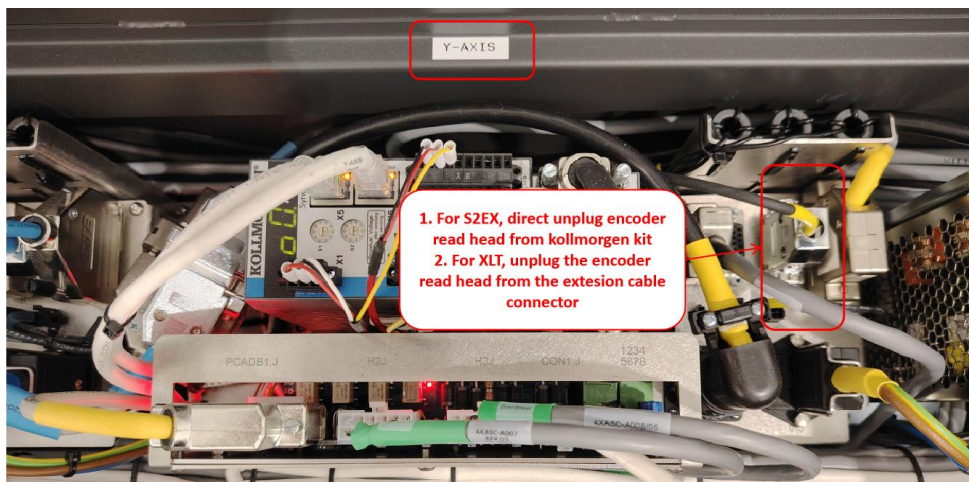


Figure 8:Y-Axis Kollmorgen Kit

b) For S3

Note: Please unplug the encoder read head from the kollmorgen kit.

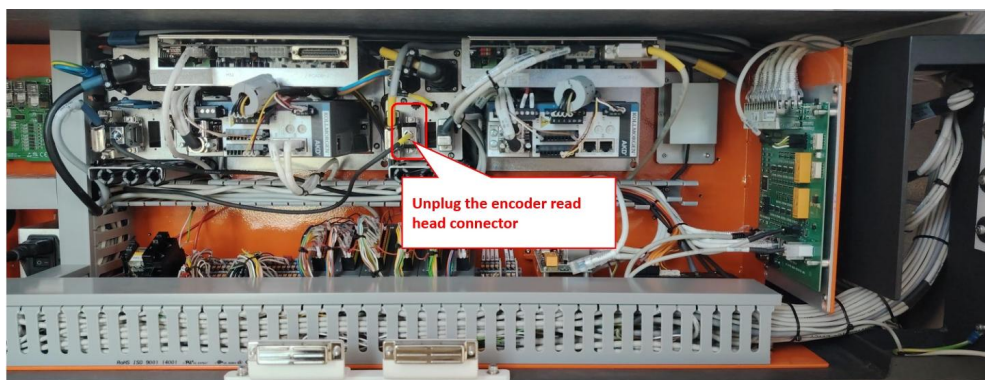


Figure 9:Y-Axis Kollmorgen Kit

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12. **Removal of cable tie and the Y-Axis Encoder Bracket:** Remove the cable tie. Then, unfasten 2*M5x20 screws to remove the Y-Axis Read Head Encoder bracket. Take caution when removing the bracket to prevent damage to the read head encoder.

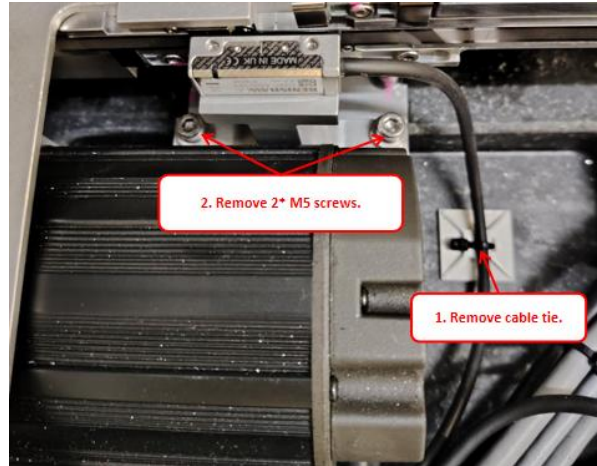


Figure 10:Y-Axis Read Head Encoder

13. **Dismantle the Read Head Encoder from the Bracket:** Unfasten the 2*M3x20 screws to remove the Read Head Encoder from the bracket.

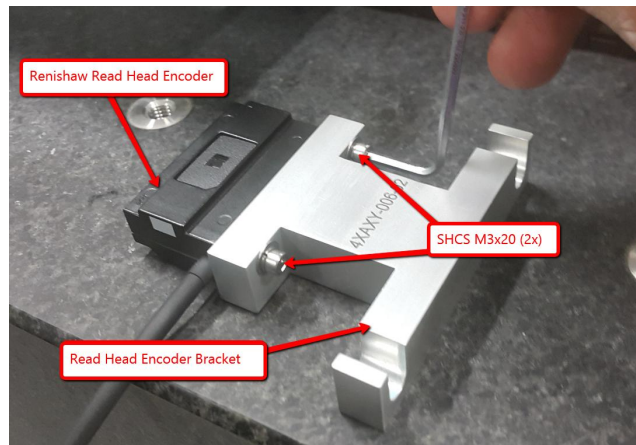


Figure 11:Y-Axis Read Head Encoder with bracket (Sample Image)

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14. **Removal of Y-Axis Read Head Encoder:** Remove Y-Axis Read Head Encoder by referring to the routing below.

Note: Follow the red arrow for removal direction, and remove any necessary cable ties.

a) For S2EX (Part ID: 4JASC-A081) and XXL/XLT (Part ID: 4XASP-A096)

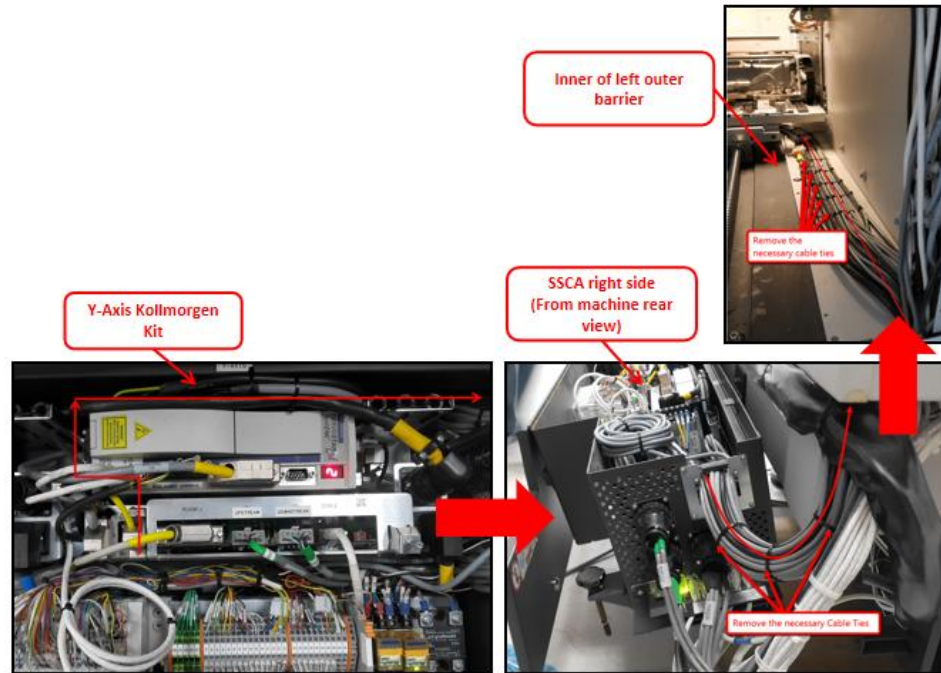


Figure 12:Y-Axis Read Head Encoder cable routing (Sample Image)

b) For S3 (Part ID: 4FASCC-A081)

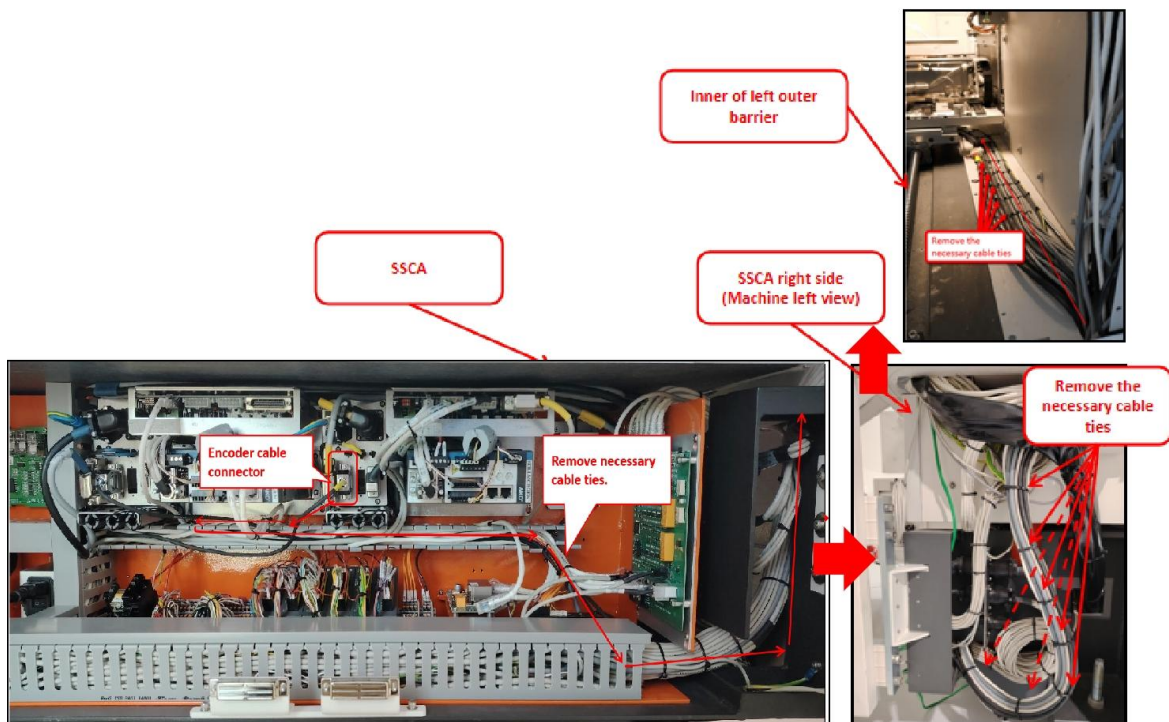


Figure 13:Y-Axis Read Head Encoder cable routing

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Origin Y-Axis Glass Scale Removal Procedure

15. **Dismantle the home sensor and limit sensor:** Unscrew the M3 screw for home sensor and limit sensor and remove it from the machine.

- a) S2EX machine (Part ID: 4XAXY-056 & 4XAXY-058)
- b) XXL / XLT machine (Part ID: 4XAXY-056 & 4XAXY-058)
- c) S3 machine (Part ID: 4FAHWP-A156 & 4FAXYS-A111)

16. **Removal of Y-Axis Cover Plate:** Unfasten the 4*M3 screws for Y-Axis cover plate.

Note: If applicable



Figure 14:Y-Axis Cover Plate

17. **Loosen the Screws for Spring Clamp:** Loosen the screws for the spring clamp with M3 Allen key.

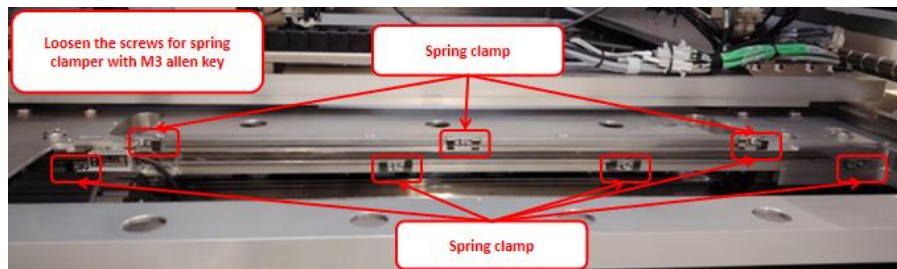


Figure 15:Y-Axis Glass Scale

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18. **Removal of Quantic Scale Top Mount Bar and Limit Sensor:** Unfasten the screws for top mount bar. Carefully remove and place it on the clean and flat surface.

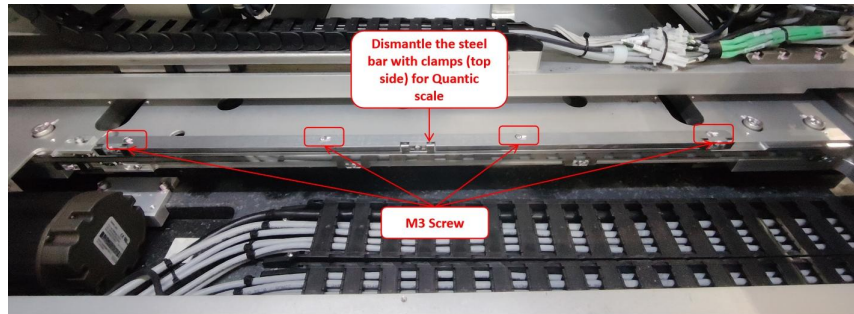


Figure 16: Y-Axis Mount Bar with Clamp (Top Side) & Limit Sensor

19. **Removal of Y-Axis Glass Scale:** Carefully remove the Y-Axis Glass scale. Take caution to place the scale in a flat surface to prevent any incidents.



Figure 17: Glass Scale

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New Y-Axis Quantic Scale Replacement

20. **Spare Part Preparation:** Refer to the table below to gather and prepare the new spare part required for the replacement process.

a) For S2EX

Part number	Part Description	QTY	UOM	illustration
4XASPP-A431	S2EX Y-AXIS QUANTIC ENCODER SCALE ASSEMBLY KIT	1	PC	
4JAXYS-A603	S2EX HOME SWITCH ASSEMBLY, Y-AXIS	1	PC	
4JAXYS-A605	S2EX LIMIT SWITCH ASSEMBLY, Y-AXIS	1	PC	

Table 3: Y-Axis Quantic Scale List for S2EX

b) For XXL / XLT

Part number	Part Description	QTY	UOM	illustration
4XASPP-A433	XLT Y-AXIS QUANTIC ENCODER SCALE ASSEMBLY KIT	1	PC	
4CAXYS-A603	XLT HOME SWITCH ASSEMBLY, Y-AXIS	1	PC	
4CAXYS-A605	XLT LIMIT SWITCH ASSEMBLY, Y-AXIS	1	PC	

Table 4: Y-Axis Quantic Scale List for XXL/XLT

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c) For S3

Part number	Part Description	QTY	UOM	illustration
4XASPP-A435	S3 Y-AXIS QUANTIC ENCODER SCALE ASSEMBLY KIT	1	PC	
4FAXYS-A603	S3 HOME SWITCH ASSEMBLY, Y-AXIS	1	PC	
4FAXYS-A605	S3 LIMIT SWITCH ASSEMBLY, Y-AXIS	1	PC	

Table 5: Y-Axis Quantic Scale List for S3

21. **Y-Axis Quantic Scale Assembly:** Assemble the Quantic scale and allow it to rest on the spring clamps. Be cautions for any obstacle and not damaging the Quantic scale.

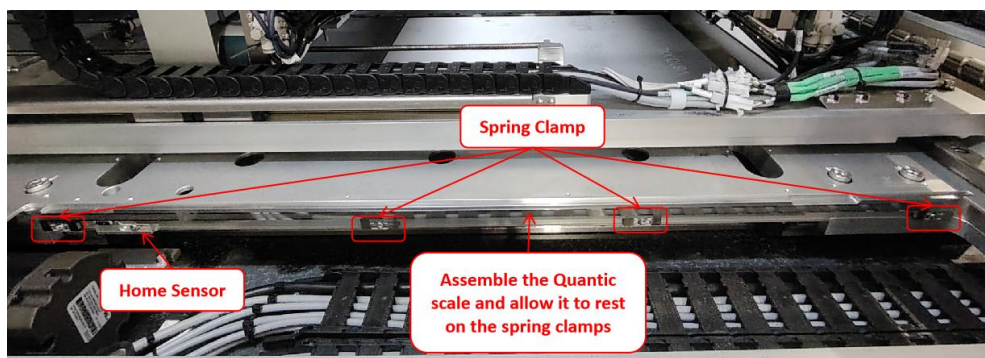


Figure 18:Y-Axis Quantic Scale (Sample Image)

22. **Steel Bar with Clamps (Top side) Assembly:** Assemble the steel bar with clamps that guiding the top side of the Quantic scale. Tighten the M3 screws after align the steel bar in position.

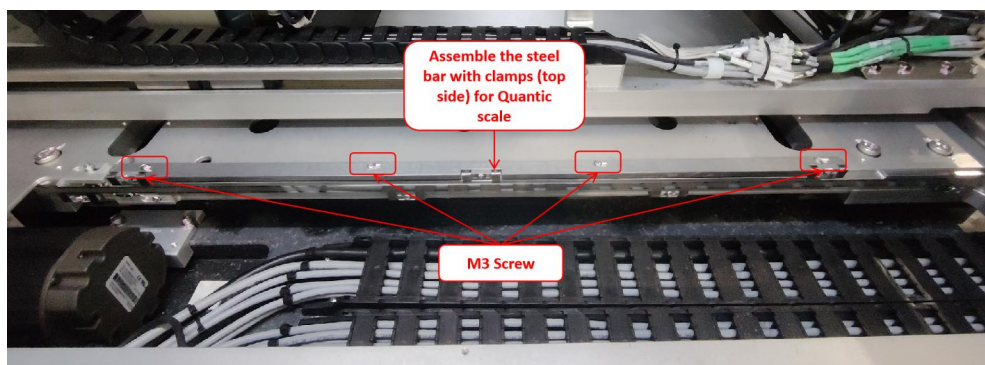


Figure 19:Y-Axis Encoder Cover Plate and Limit Sensor (Sample Image)

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23. **Home and Limit Sensors Assembly:** Align the home sensor and tighten the screw in the center position of the elongate. Then, assemble the limit sensor and tighten the M3 screw.

- S2EX machine (Part ID: 4JAXYS-A603 & 4JAXYS-A605)
- XXL / XLT machine (Part ID: 4CAXYS-A603 & 4CAXYS-A605)
- S3 machine (Part ID: 4FAXYS-A603 & 4FAXYS-A605)

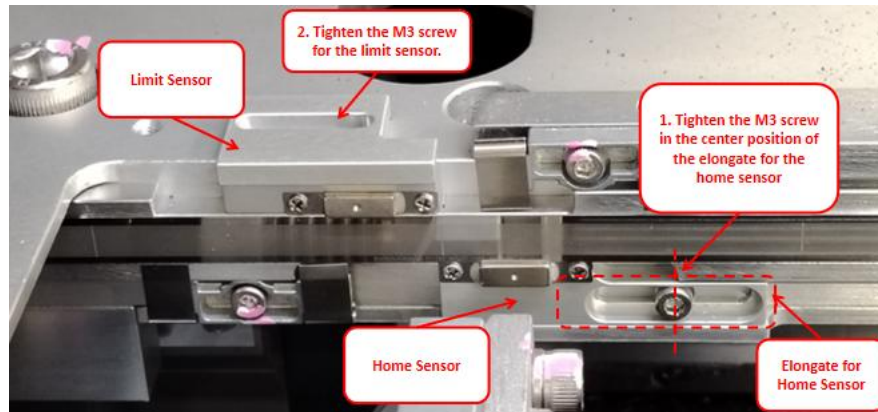


Figure 20: Y-Axis Home and Limit Sensors Alignment

24. **Replace New Quantic Glass Scale:** Proceed to replace new Quantic glass scale by referring to “**New Y-Axis Encoder Read Head Replacement**” procedure.

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New Y-Axis Encoder Read Head Replacement

25. **Spare Part Preparation:** Refer to the table below to gather and prepare the new spare part required for the replacement process.

a) For S2EX

Part number	Part Description	QTY	UOM	illustration
2675-0053	Encoder Head, Optical, 0.5UM	1	PC	
4JAXYS-N476-01	ENCODER READHEAD BRACKET	1	PC	
2720-0092	SOCKET HEAD SCREW,HEX,S/S,M2.5,20	2	PC	
2760-0001	SPRING WASHER,S/S,M2.5	2	PC	
2760-0035	FLAT WASHER,S/S,M2.5	2	PC	

Table 6: Spare Part List for S2EX

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b) For XXL / XLT

Part number	Part Description	QTY	UOM	illustration
2675-0053	Encoder Head, Optical, 0.5UM	1	PC	
4CAXYS-N287-02	XLT, Y-AXIS ENCODER BRACKET	1	PC	
2720-0092	SOCKET HEAD SCREW,HEX,S/S,M2.5,20	2	PC	
2760-0001	SPRING WASHER,S/S,M2.5	2	PC	
2760-0035	FLAT WASHER,S/S,M2.5	2	PC	
4CASCC-A260	XLT, LINEAR ENCODER Y-AXIS EXTENSION	1	PC	

Table 7: Spare Part List for XXL/XLT

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c) For S3

Part number	Part Description	QTY	UOM	illustration
2675-0053	Encoder Head, Optical, 0.5UM	1	PC	
4FAXYS-N278-00	S3, Y-AXIS ENCODER BRACKET	1	PC	
2720-0092	SOCKET HEAD SCREW,HEX,S/S,M2.5,20	2	PC	
2760-0001	SPRING WASHER,S/S,M2.5	2	PC	
2760-0035	FLAT WASHER,S/S,M2.5	2	PC	
4FASCC-A260	S3, LINEAR ENCODER Y-AXIS EXTENSION	1	PC	

Table 8: Spare Part List for S3

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26. Read Head Encoder and Bracket Assembly: Assemble new Y-Axis Read Head Encoder to the bracket and tighten the screws.

- a) S2EX machine (Part ID: 2675-0053, 4JAXYS-N476-01, 2720-0092*2, 2760-0001*2, 2760-0035*2)
- b) XXL / XLT machine (Part ID: 2675-0053, 4CAXYS-N287-02, 2720-0092*2, 2760-0001*2, 2760-0035*2)
- c) S3 machine (Part ID: 2675-0053, 4FAXYS-N278-00, 2720-0092*2, 2760-0001*2, 2760-0035*2)

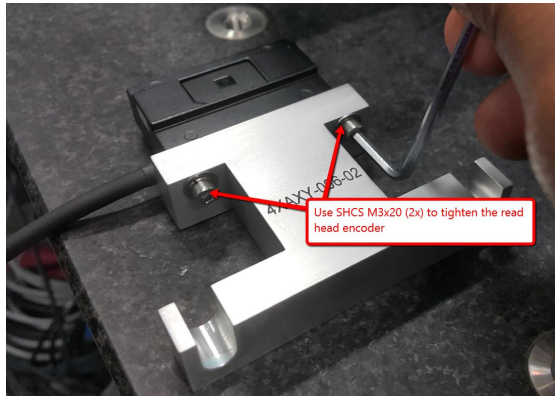


Figure 21: Y-Axis Read Head Encoder with bracket (Sample Image)

27. Read Head Encoder Bracket Assembly: Install the Read Head Encoder bracket on the machine and tighten the 2*M5x20 screws. Then, apply a cable tie to avoid cable damage during motion.

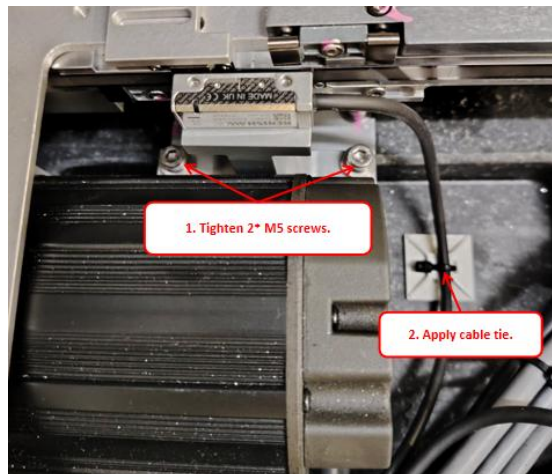


Figure 22: Y-Axis Read Head Encoder Bracket with Encoder Head

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28. **Cable Routing:** Route the cable following the impacted area numbering in sequence.

Note: the cable should route thru underneath of servo motor.

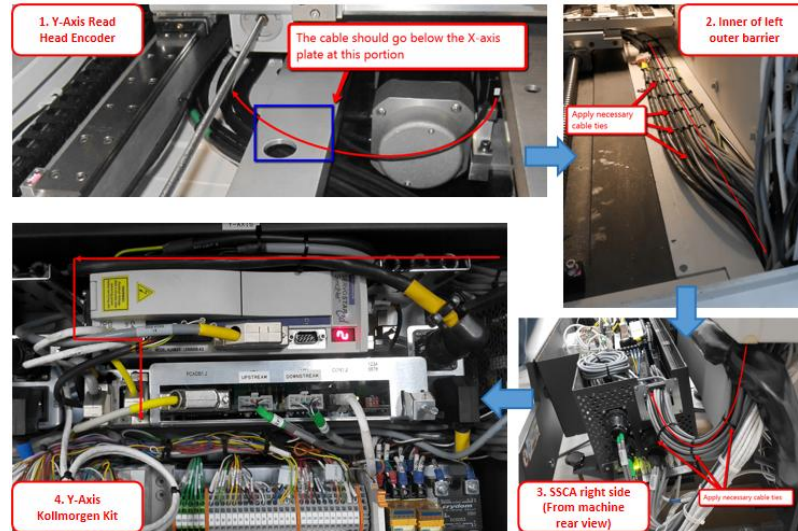


Figure 23: Y-Axis Read Head Encoder Cable Routing (Sample Image)

29. **Read Head Encoder Connection with Extension Cable:** Connect the read head encoder with the extension cable to the SSCA to restore full functionality. Ensure proper wiring management for the extension cable. Refer extension cable part ID per following.

- a) XXL/XLT machine (Part ID: 4CASC-A260)
- b) S3 machine (Part ID: 4FASCC-A260)

30. **Re-connect the Y-Axis Read Head Encoder Connector:** Plug in the Y-Axis Read Head Encoder connector in the Y-Axis Kollmorgen kit or to the extension cable.

- a) For S2EX and XXL/XLT

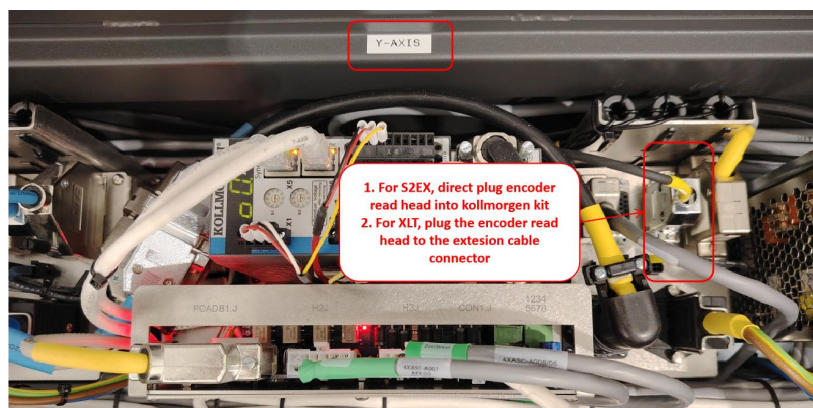


Figure 24: Y-Axis Kollmorgen Kit

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b) For S3

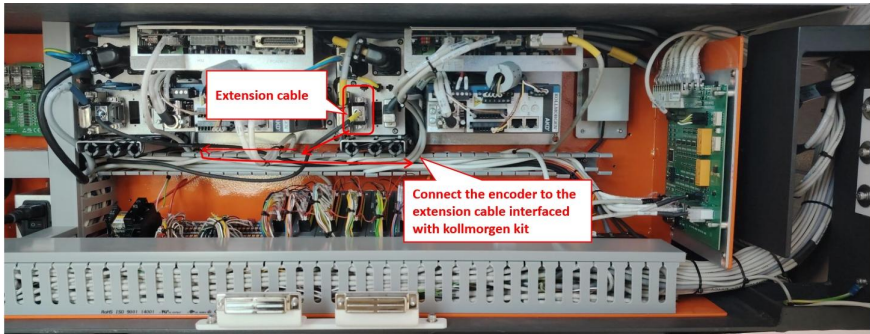


Figure 25: Y-Axis Kollmorgen Kit

31. Cable Clamp Assembly: Place the cable clamp on its position and tighten back the 2 screws.

a) For S2EX and XXL/XLT

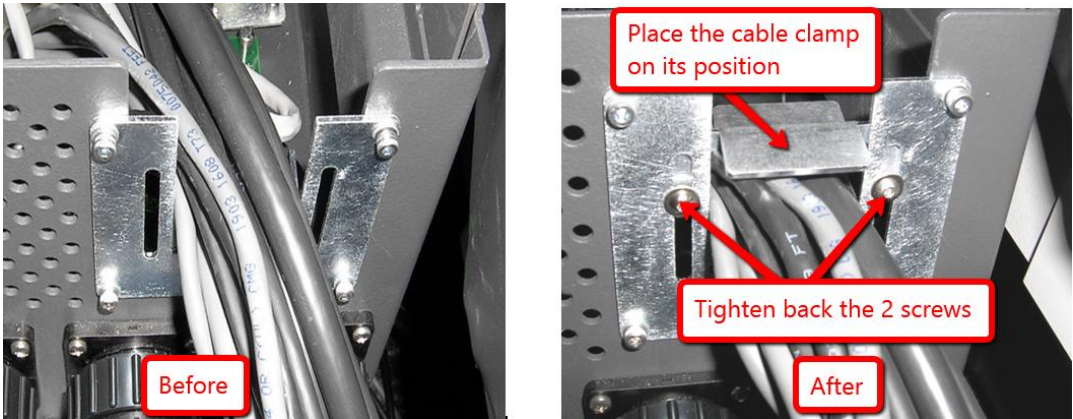


Figure 26: SSCA (Sample Image)

b) For S3

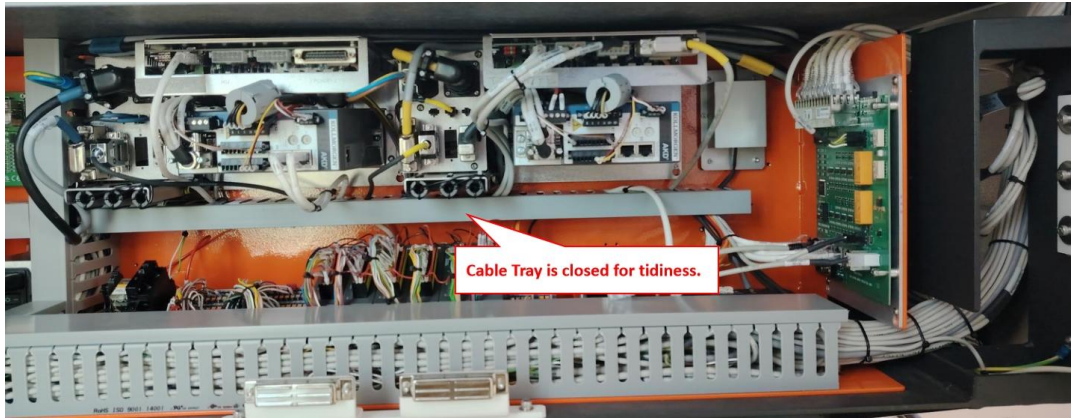


Figure 27: SSCA

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32. **Horizontal Lead Cover Assembly:** Place back the horizontal lead cover and push it in the direction as shown in below figure. Then, tighten back the 3*M5 screws.

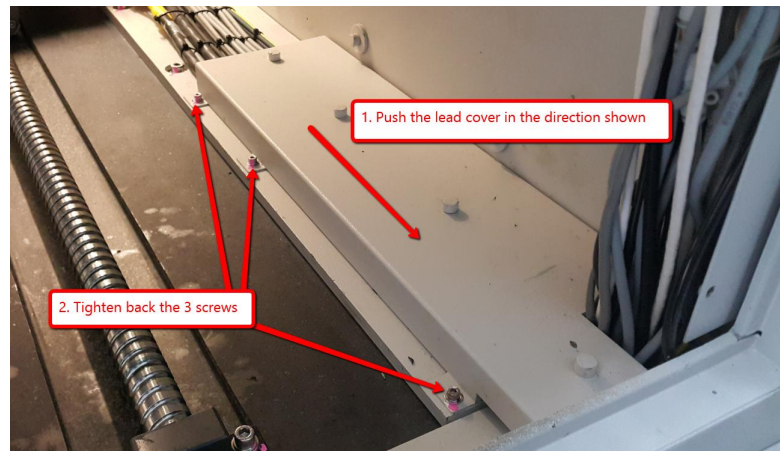


Figure 28: Horizontal Lead Cover (Sample Image)

33. **Vertical Lead Cover Assembly:** Assemble the vertical lead cover and tighten the screws.

a) For S2EX and XXL/XLT



Figure 29: Vertical Lead Cover (Sample Image)

b) For S3

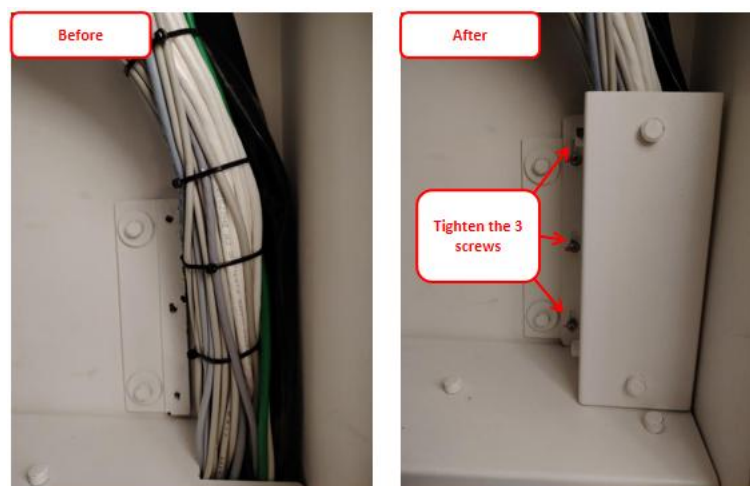


Figure 30: Vertical Lead Cover

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Tool Removal and Cleaning

34. Before concluding the replacement procedure, carefully inspect the machine interior to ensure that all tools, equipment, and debris have been removed. Retrieve any tools or equipment left inside the machine and return them to their designated storage area to prevent any interference with machine operation.
35. Dispose of any removed cable ties, packaging materials, or other debris in the designated waste disposal containers according to company guidelines or environmental regulations.
36. Once the interior surfaces have been cleaned and cleared of debris, visually inspect the machine to ensure that it is in a tidy and organized state, ready for operation.
37. Perform a final check to verify that all components have been properly installed and secured according to manufacturer specifications. Confirm that all cables are properly routed and secured within their respective E-chains to prevent any interference with machine movement.
38. Once the machine has been cleaned and inspected, power on the system according to standard operating procedures but without launch the V810 GUI.

Reference for Alignment Procedure

39. **Y-Axis Encoder Read Head Alignment:** Refer to “AXI V810 Quantic Encoder Read Head Alignment Procedure”. to verify the encoder status and perform necessary alignment if needed.

Note: Crucial to ensure the signal strength for encoder read head along the y-axis stroke.

40. **Home and Limit Sensors Alignment:** After validating the encoder head performance, refer to **AXI V810 Quantic Home and Limit Sensors Upgrade Alignment Procedure (AKD, S200, Panasonic)** and perform necessary alignment and adjustment.

Note: It's a must be performed whenever there is any adjustment or replacement of Quantic scale.